

# **POINTE-AU-PERE, QC, Canada**

WMO: 715540

Lat: 48.5139N

Lon: 68.4683W

Elev: 16

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 99999

## **Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -7.2       | -2.5 | -16.3                         | 2.3 | -6.4 | -11.5 | 3.0 | -1.7 | 35.2                  | 17.9 | 31.9 | 16.4 | 13.9                  | 290  | 0.718 |

## **Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 14.8                   | 73.6            | 65.5 | 70.8 | 63.8 | 68.4 | 62.0 | 67.3                | 72.1 | 65.0 | 69.6 | 62.9 | 67.4 | 11.1                 | 200  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 65.3                            | 93.4 | 70.7 | 62.9 | 85.7 | 68.3 | 60.8 | 79.7 | 66.1 | 31.9          | 72.4 | 30.1 | 69.8 | 28.5 | 67.5 | 77.0              |

## **Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |     |                    | n-Year Return Period Values of Extreme Temperature |           |       |            |       |            |       |            |     |
|-------------------|------|------|----------------------------|------|-----|--------------------|----------------------------------------------------|-----------|-------|------------|-------|------------|-------|------------|-----|
| 1%                |      | 2.5% | 5%                         | Mean |     | Standard Deviation |                                                    | n=5 years |       | n=10 years |       | n=20 years |       | n=50 years |     |
| Min               | Max  | Min  | Max                        | Min  | Max | Min                | Max                                                | Min       | Max   | Min        | Max   | Min        | Max   | Min        | Max |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g) | (h)                | (i)                                                | (j)       | (k)   | (l)        | (m)   | (n)        | (o)   | (p)        |     |
| 28.7              | 25.1 | 22.2 | -13.2                      | 83.3 | 5.8 | 4.3                | -17.3                                              | 86.3      | -20.7 | 88.8       | -24.0 | 91.2       | -28.2 | 94.3       | DB  |
|                   |      |      | -13.6                      | 72.7 | 5.6 | 2.2                | -17.7                                              | 74.3      | -20.9 | 75.5       | -24.0 | 76.8       | -28.1 | 78.3       |     |

## **Monthly Climatic Design Conditions**

|                                                                                  |                                                                                  | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                                  |                                                                                  | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg                                                                            | 38.9   | 14.3  | 16.2  | 24.6  | 36.2  | 46.8  | 55.0  | 60.4  | 59.3  | 52.9  | 43.3  | 33.6  | 22.3 |      |
|                                                                                  | DBStd                                                                            | 17.49  | 10.65 | 9.56  | 8.82  | 5.99  | 5.86  | 5.60  | 4.34  | 4.35  | 5.54  | 6.04  | 7.45  | 8.99 |      |
|                                                                                  | HDD50                                                                            | 5016   | 1106  | 947   | 788   | 415   | 135   | 15    | 0     | 0     | 31    | 226   | 493   | 860  |      |
|                                                                                  | HDD65                                                                            | 9561   | 1571  | 1367  | 1253  | 864   | 564   | 303   | 153   | 183   | 363   | 674   | 941   | 1325 |      |
|                                                                                  | CDD50                                                                            | 953    | 0     | 0     | 0     | 1     | 37    | 166   | 322   | 289   | 119   | 17    | 2     | 0    |      |
|                                                                                  | CDD65                                                                            | 21     | 0     | 0     | 0     | 0     | 0     | 3     | 10    | 7     | 1     | 0     | 0     | 0    |      |
|                                                                                  | CDH74                                                                            | 90     | 0     | 0     | 0     | 0     | 3     | 17    | 39    | 28    | 3     | 0     | 0     | 0    |      |
|                                                                                  | CDH80                                                                            | 12     | 0     | 0     | 0     | 0     | 0     | 2     | 5     | 5     | 0     | 0     | 0     | 0    |      |
| Wind                                                                             | WSAvg                                                                            | 10.0   | 12.3  | 11.4  | 11.4  | 10.3  | 8.7   | 7.9   | 7.7   | 7.6   | 8.6   | 10.3  | 11.8  | 12.3 |      |
| Precipitation                                                                    | PrecAvg                                                                          | 37.3   | 2.2   | 2.4   | 2.8   | 2.7   | 3.2   | 3.5   | 3.5   | 3.3   | 3.6   | 3.6   | 3.0   | 3.8  |      |
|                                                                                  | PrecMax                                                                          | 43.7   | 4.0   | 4.4   | 5.3   | 5.0   | 5.6   | 6.8   | 6.0   | 6.2   | 10.0  | 5.6   | 6.1   | 5.3  |      |
|                                                                                  | PrecMin                                                                          | 27.4   | 0.9   | 1.1   | 1.7   | 1.3   | 1.8   | 1.3   | 0.7   | 1.0   | 0.9   | 1.3   | 1.0   | 1.9  |      |
|                                                                                  | PrecStd                                                                          | 4.3    | 0.9   | 0.9   | 1.1   | 1.1   | 1.1   | 1.4   | 1.4   | 1.4   | 2.0   | 1.3   | 1.1   | 0.9  |      |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                             | DB     | 43.9  | 40.4  | 47.0  | 57.7  | 70.8  | 76.1  | 78.3  | 77.3  | 73.2  | 66.5  | 61.2  | 44.9 |      |
|                                                                                  |                                                                                  | MCWB   | 41.5  | 37.9  | 42.4  | 49.5  | 59.2  | 65.5  | 69.8  | 67.9  | 65.6  | 58.8  | 56.1  | 42.2 |      |
|                                                                                  | 2%                                                                               | DB     | 37.6  | 35.1  | 41.3  | 50.6  | 64.4  | 71.2  | 73.6  | 72.2  | 68.1  | 60.1  | 52.1  | 39.5 |      |
|                                                                                  |                                                                                  | MCWB   | 35.7  | 33.0  | 37.5  | 43.4  | 55.3  | 62.6  | 66.3  | 65.1  | 62.0  | 54.7  | 47.5  | 37.7 |      |
|                                                                                  | 5%                                                                               | DB     | 33.5  | 32.7  | 38.5  | 47.0  | 59.9  | 67.4  | 70.9  | 69.4  | 64.6  | 55.3  | 46.1  | 36.5 |      |
|                                                                                  |                                                                                  | MCWB   | 31.9  | 31.1  | 35.2  | 41.2  | 52.1  | 60.1  | 64.4  | 63.3  | 59.4  | 50.8  | 42.7  | 34.5 |      |
|                                                                                  | 10%                                                                              | DB     | 29.5  | 30.4  | 36.1  | 44.1  | 56.1  | 64.4  | 68.4  | 67.0  | 61.7  | 51.8  | 42.6  | 34.1 |      |
|                                                                                  |                                                                                  | MCWB   | 27.7  | 28.8  | 33.3  | 39.3  | 49.7  | 58.1  | 62.5  | 61.7  | 57.1  | 48.1  | 39.9  | 32.3 |      |
|                                                                                  | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 41.5  | 38.4  | 43.0  | 50.0  | 61.7  | 68.5  | 71.2  | 69.6  | 66.7  | 59.9  | 58.1 | 43.0 |
|                                                                                  |                                                                                  |        | MCDB  | 43.5  | 40.3  | 46.2  | 54.8  | 68.7  | 74.6  | 76.7  | 74.8  | 72.0  | 64.6  | 63.5 | 44.9 |
| 2%                                                                               |                                                                                  | WB     | 35.8  | 33.5  | 38.1  | 44.6  | 56.8  | 63.9  | 67.8  | 66.5  | 63.0  | 54.9  | 48.2  | 38.0 |      |
|                                                                                  |                                                                                  | MCDB   | 37.2  | 34.7  | 40.5  | 49.2  | 62.5  | 69.4  | 72.4  | 70.7  | 67.0  | 59.3  | 51.7  | 39.4 |      |
| 5%                                                                               |                                                                                  | WB     | 32.1  | 31.3  | 35.6  | 42.0  | 53.1  | 61.2  | 65.4  | 64.2  | 59.8  | 51.5  | 43.1  | 34.8 |      |
|                                                                                  |                                                                                  | MCDB   | 33.6  | 32.6  | 38.1  | 46.2  | 58.9  | 66.3  | 69.9  | 68.5  | 64.0  | 54.8  | 45.6  | 36.2 |      |
| 10%                                                                              |                                                                                  | WB     | 27.9  | 28.9  | 33.5  | 39.7  | 50.4  | 58.7  | 63.2  | 62.2  | 57.4  | 48.4  | 40.3  | 32.6 |      |
|                                                                                  |                                                                                  | MCDB   | 29.5  | 30.4  | 35.6  | 43.2  | 55.4  | 63.5  | 67.6  | 66.2  | 61.2  | 51.3  | 42.3  | 33.9 |      |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB                                                                            | MDBR   | 12.6  | 12.7  | 12.0  | 11.2  | 13.5  | 14.7  | 14.8  | 14.3  | 13.2  | 10.4  | 9.0   | 10.2 |      |
|                                                                                  |                                                                                  | MCDBR  | 15.2  | 12.4  | 13.1  | 16.8  | 20.7  | 19.3  | 18.4  | 17.7  | 17.7  | 16.5  | 14.1  | 12.2 |      |
|                                                                                  |                                                                                  | MCWBR  | 14.9  | 11.7  | 11.1  | 12.0  | 13.5  | 12.8  | 12.7  | 12.0  | 13.2  | 12.5  | 11.9  | 11.9 |      |
|                                                                                  | 5% WB                                                                            | MCDBR  | 14.9  | 11.9  | 12.4  | 15.7  | 19.3  | 18.3  | 17.9  | 17.0  | 17.2  | 15.8  | 13.3  | 12.2 |      |
|                                                                                  |                                                                                  | MCWBR  | 14.7  | 11.5  | 11.0  | 11.9  | 13.7  | 13.4  | 13.6  | 12.7  | 13.6  | 12.6  | 11.7  | 12.1 |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub                                                                             | 0.264  | 0.273 | 0.303 | 0.357 | 0.367 | 0.389 | 0.413 | 0.393 | 0.367 | 0.330 | 0.299 | 0.271 |      |      |
|                                                                                  | taud                                                                             | 2.346  | 2.315 | 2.292 | 2.297 | 2.347 | 2.351 | 2.317 | 2.374 | 2.428 | 2.513 | 2.460 | 2.379 |      |      |
|                                                                                  | Ebn at Noon                                                                      | 258    | 282   | 290   | 283   | 284   | 277   | 268   | 269   | 265   | 259   | 243   | 239   |      |      |
|                                                                                  | Edh at Noon                                                                      | 22     | 28    | 34    | 38    | 38    | 38    | 39    | 35    | 30    | 23    | 20    | 19    |      |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg                                                                           | 398    | 691   | 1053  | 1391  | 1639  | 1745  | 1721  | 1472  | 1131  | 664   | 358   | 267   |      |      |
|                                                                                  | RadStd                                                                           | 40     | 76    | 78    | 120   | 152   | 128   | 169   | 96    | 74    | 53    | 36    | 24    |      |      |

## **Historical Trends**

|      |                         | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (25 neighbors) | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page